PROJECT DESIGN PHASE

Electric Vehicle Market Performance & Insights

Dashboard, KPI cards, worksheets, and Tableau Story design

1. Design Objective

This phase documents the design of the completed EV market analytics experience in Tableau Public. It explains the purpose of the KPI cards and worksheets, the dashboard structure, and the sequence used in the Tableau Story. The design is intended to let viewers move from a high-level summary to focused comparisons of sales, manufacturers, regions, categories, and vehicle specifications.

2. KPI Cards Overview

The KPI cards provide a quick summary before users explore the detailed charts. The measure definitions below reflect the fields used for the project; each KPI should be interpreted with its field units and aggregation in mind.

KPI Card	Measure / Calculation	Purpose
Total EV Models	COUNTD(vehicle_id)	Counts distinct vehicle IDs represented in the dataset.
Annual Sales	SUM(annual_sales_units)	Summarizes the annual sales units represented by the selected data and filters.
Average EV Price	AVG(price_usd)	Shows the average listed price in USD for the records in the current view.
Average Battery Capacity	Average of the battery-capacity field, using the dataset's recorded unit	Summarizes battery capacity; confirm the exact field name and unit in the data source.
Average Driving Range	AVG(range_km)	Shows average driving range in kilometres for the records in the current view.
Average Customer Rating	Average of the customer-rating field, if populated and consistently defined	Provides a high-level rating summary where the field is available and meaningful.

KPI interpretation note: KPI values respond to the current data and filters. "Total EV Models" counts distinct vehicle IDs, not necessarily unique commercial model names. Average values can be affected by missing data and should not be compared across groups without checking the underlying records.

3. Worksheet Overview

The following worksheets were used in the dashboard and/or Tableau Story.

Worksheet	Chart / View	Question It Helps Answer
Sales Trend	Line chart	How does the sales measure change over the available time period?
Top 10 Manufacturers	Ranked bar chart	Which ten manufacturers rank highest by the selected sales or market measure?
Region vs Category	Heatmap / highlight table	How does the selected measure vary across regions and EV categories?
Manufacturer Treemap	Treemap	How is the selected measure distributed across manufacturers?

Category-wise Sale	Donut chart	What share of the selected sales measure belongs to each vehicle category?
EV Distribution Map	Geographic map	Where are the records or sales distributed, based on the geographic fields available?
Capacity vs Range	Scatter plot	Is there a visible relationship between battery capacity and driving range, and are there outliers?
Battery Tech Comp	Pie chart	How is the dataset distributed across the battery technology groups?
Price Distribution	Box plot	How does price vary across the selected categories, including spread and potential outliers?

4. Dashboard Design

- The dashboard begins with a clear project title and a row of KPI cards to establish the overall context.
- Sales Trend and Top 10 Manufacturers support time-based and ranking comparisons near the top of the analytical flow.
- Region vs Category, Manufacturer Treemap, Category-wise Sale, and EV Distribution Map provide complementary views of market composition and geography.
- Capacity vs Range, Battery Tech Comp, and Price Distribution focus on technical attributes and price variation.
- The chart mix combines comparison, composition, geographic, and relationship views so that one chart type is not used for every question.
- Filters and tooltips should make the views easier to explore while keeping titles, labels, and units understandable.

5. Tableau Story Structure

The Tableau Story uses the dashboard and individual worksheets as a guided sequence. The story sheets were set to Entire View so that each view fits within the available story-point area. After Price Distribution, a final dashboard has been added to bring the analysis together through written key insights, strategic recommendations, and a conclusion. This closing point turns the visual exploration into a concise, decision-oriented summary.

Story Point / Sheet	Role in the Narrative	Insight to Explore
1. Dashboard	Introduce the overall EV market view and KPI summary.	What does the dataset show at a glance?
2. Sales Trend	Explore change across the available time period.	Are there increases, decreases, or fluctuations in the observed sales measure?
3. Top 10 Manufacturers	Compare the leading manufacturers.	Which manufacturers rank highest, and how large are the differences between them?
4. Region vs Category	Compare regional patterns across vehicle categories.	Do category distributions appear similar or different across regions?
5. Manufacturer Treemap	Show relative manufacturer contribution.	Are records or sales concentrated among a few manufacturers or spread across many?
6. Category-wise Sale	Compare the share of sales across categories.	Which categories account for the largest and smallest shares in the current view?
7. EV Distribution Map	Explore geographic distribution.	Where do the records or sales appear concentrated, based on the mapped field?
8. Capacity vs Range	Explore a technical relationship.	Does higher battery capacity tend to coincide with longer range in this dataset, and where are exceptions?
9. Battery Tech Comp	Summarize battery technology	Which technology groups are most

	composition.	represented in the data?
10. Price Distribution	Compare price spread and outliers.	Which categories have higher or lower typical prices and wider variation?
11. Key Insights, Strategic Recommendations & Conclusion Dashboard	Close the Story with a text-based synthesis of the analysis, summarizing the key insights, practical strategic recommendations, and overall conclusion.	What are the main takeaways from the charts, what actions could stakeholders consider, and what is the overall conclusion supported by the dataset?

6. Dashboard and Story Insights

The dashboard and Story are designed to surface the following types of insight. The final Key Insights, Strategic Recommendations & Conclusion dashboard follows Price Distribution and synthesizes the findings into readable text. Its statements and recommendations should be grounded in the patterns actually observed in Tableau. These are the analytical questions and comparisons supported by the completed views, rather than claims of specific numerical results.

- **Market performance over time:** The Sales Trend view helps reveal periods of growth, decline, or fluctuation in the sales measure, provided the time field is valid.
- **Manufacturer concentration:** The Top 10 Manufacturers and Manufacturer Treemap views show whether the selected measure is concentrated among a small group or more evenly distributed.
- **Regional and category differences:** The Region vs Category heatmap helps compare how categories vary across regions and identify combinations that merit closer review.
- **Category contribution:** The Category-wise Sale donut chart compares each category's share of the selected sales measure, making larger and smaller contributions easier to see.
- **Geographic distribution:** The EV Distribution Map reveals the locations represented most strongly in the data, subject to the quality and meaning of the geographic field.
- **Battery capacity and range:** The Capacity vs Range scatter plot allows users to inspect the direction and strength of any visible relationship and identify unusual records. A visible association does not prove causation.
- **Battery technology mix:** Battery Tech Comp summarizes the representation of each technology group in the dataset; representation should not automatically be interpreted as market share unless the data supports that meaning.
- **Price variation:** Price Distribution compares typical price levels, spread, and potential outliers across the chosen grouping, helping show where prices are more or less variable.

7. Visual and Interaction Guidelines

- Use a consistent colour palette, typography, spacing, and number formatting across the worksheets.
- Keep chart titles specific to the measure and grouping shown.
- Display units for price, battery capacity, range, and sales wherever applicable.
- Ensure filters and dashboard actions behave consistently across relevant views.
- Use tooltips to provide additional detail without crowding the main chart.
- Avoid interpreting counts as sales or market share unless the metric definition supports that interpretation.

8. Design Assumptions and Validation Notes

This document describes the worksheets and KPI definitions recorded for the project. Before treating any observation as a final finding, check the published dashboard values and confirm the exact field names, units, aggregation, filter state, and data coverage. The combined train/test dataset was prepared for exploratory visualization; any conclusions should be framed as patterns in the supplied data, not as proof of causation or a guarantee of future EV market behaviour. No numerical ranking, percentage, trend direction, or correlation strength is asserted here because those exact displayed results have not been recorded in this design document.

9. Design Phase Deliverables

The design deliverables are the KPI card definitions, worksheet inventory, dashboard structure, Tableau Story sequence, and the analytical questions that each view is intended to answer. These provide the design reference for project development and subsequent testing.